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SHORT-PAPER

Generative Next-Basket Recommendation

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ABSTRACT

Next-basket Recommendation (NBR) refers to the task of predicting a set of items that a user will purchase in the next basket. However, most of existing works merely focus on the correlations between user preferences and predicted items, ignoring the essential correlations among items in the next basket, which often results in over-homogenization of predicted items. In this work, we presents a **Gen**erative next-basket **Rec**ommendation model (**GenRec**), a novel NBR paradigm that generates the recommended items one by one to form the next basket via an autoregressive decoder. This generative NBR paradigm contributes to capturing and considering item correlations inside each baskets in both training and serving. Moreover, we jointly consider user's both item- and basket-level contextual information to better capture user's multi-granularity preferences. Extensive experiments on three real-world datasets demonstrate the effectiveness of our model.

CCS CONCEPTS

• **Information systems → Recommender systems**.

KEYWORDS

Next-basket recommendation, autoregressive generative model.

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1 INTRODUCTION

Nowadays, recommender systems (RS) have become increasingly important to help users discover relevant items of interest in real-world applications. As a fundamental recommendation tasks, next-basket recommendation (NBR) aims to recommend a basket (*i.e.,* a set of items) to a user based on a sequence of his/her historical purchased baskets, which is particularly prevalent in e-commerce and retail industry [11, 13, 14, 20, 23, 28]. In NBR, purchased items in a basket are usually relevant (*e.g.,* a soccer beginner will buy *football shoes*, *jerseys*, *sports drink* in a basket). Therefore, a good next-basket recommender needs to consider the correlations among the recommended items in prediction.

Figure 1: (a) An illustrative issue of over-homogenization in most existing deep learning based NBR models. (b) Our generative sequential model considers the correlations of items inside the generated basket, which could address the over-homogenization issue.

As early attempts, Markov Chains (MC) based methods (*e.g.,* FPMC [22]) have been borrowed to model the basket sequences, while most of these studies focus on learning the item-item transitions to predict the next basket, which is insufficient to capture users' long-term preference in basket sequences. HRM [28] leverages a two-layer architecture to construct a no-linear hybrid aggregation of the user vector and the last transaction representation. In recent years, deep learning based NBR methods [1, 2, 4, 20, 25, 34, 35] have been proposed to model long- and short-term user preferences. The basic idea is to leverage Recurrent Neural Networks (RNN) or other deep learning techniques to build global sequential

features which reflect interactions among baskets, and represent user's dynamic interests over time. Specifically, DREAM [34] leverages RNNs to consider both user's dynamic interest and global sequential features that reflect interactions among baskets. CLEA [20] denoises irrelevant items in the history with contrastive learning to predict next basket. In another line of work, personalized item frequency (PIF) is used in NBR recently [7, 10, 11, 17], which is based on the repeated purchased behaviors and collaborative purchased patterns widely existing in users' purchased records. In particular, TIFU-KNN [11] leverages a nearest neighbor-based model for NBR that directly models the personal history of users. These methods are suitable for datasets with lots of repeated behaviors.

Despite their effectiveness, most of the existing NBR models based on deep learning essentially lack the consideration of explicitly measuring the correlations among items inside the generated baskets, as shown in Figure 1. Firstly, existing NBR methods usually recommend the top-K ranked items to form the basket according to the learned user representation. It may lead to severe over-homogenization [30, 32, 33] among the recommended items (*e.g.,* recommending multiple *shoes*), while users often purchase relevant and diversified items (*e.g., shoes* and *jerseys*) to form an appropriate combination instead of the same type of items in practice. Secondly, most of these approaches aggregate a set of items into a basket-level representation by matrix transformation, thus losing some item-level contextual information.

In order to discover and highlight possible correlations among items in next basket to meet users' real demands, we propose a new NBR paradigm **Generative next-basket Recommendation model (GenRec)**, which sequentially generates items one by one to build the basket with a flexible length through its item- and basket-level encoders and autoregressive item-level decoder, inspired by the success of generative models [5, 9, 16, 21, 27]. The generative sequential model perfectly fits the NBR task for it could capture and make full use of the implicit correlations among items in generating more accurate and diversified baskets. Moreover, we jointly consider both item- and basket-level modeling in the encoders, which completely maintains the item-level contextual information, making messages of items and baskets cooperate well to facilitate sequential modeling for better generation.

Our main contributions are summarized as follows: (1) To the best of our knowledge, it is the first NBR method which considers the correlations among items in the next basket by generating the items one by one; (2) We propose a novel generative NBR paradigm modeling both item- and basket-level contextual information to meet users' real purchase demands; (3) Extensive experiments conducted on three real-world datasets demonstrate the effectiveness of our proposed approach.

2 METHODOLOGY

In this section, we present the proposed Generative Next-Basket Recommendation approach, called GenRec. The overall illustration of the proposed approach is depicted in Figure 2.

Now, we start with a brief description of the typical NBR. Assuming that we have a set of users and items, denoted by $\mathcal{U}$ and $\mathcal{I}$ respectively. Let $u \in \mathcal{U}$ denote a user and $i \in \mathcal{I}$ denote an item. The number of users and items are donated as $|\mathcal{U}|$ and $|\mathcal{I}|$. Given a user u, his/her purchase record sorted by the timestamp (denoted as t) is a basket sequence $B^u = \{B^u_1, B^u_2, \ldots, B^u_T\}$. $B^u_t \subseteq \mathcal{I}$ $(1 \leq t \leq T)$ consists of a set of items. NBR aims to predict next basket of items in $B^u_{(T+1)}$ that u would probably buy based on his/her historical purchase.

2.1 The Proposed Generative Next-Basket Recommendation

Item Embedding. We maintain an item embedding matrix $\boldsymbol{M}_{\mathcal{I}} \in \mathbb{R}^{|\mathcal{I}| \times D}$ (the embedding dimension denoted as D) to project the high dimensional one-hot representation of each item to the low dimensional dense representation. Given a basket sequence including T baskets (with N items), we incorporate learnable position embedding matrix $\boldsymbol{P} \in \mathbb{R}^{N \times D}$ and whole-basket embedding matrix $\boldsymbol{W} \in \mathbb{R}^{T \times D}$ (*i.e.,* items within the same basket share the same whole-basket embedding) respectively to facilitate the item-level sequence representation. For a given item, its input representation $\boldsymbol{h}_i$ is constructed by summing the corresponding *item*, *position* and *whole-basket embedding* denoted as $\boldsymbol{v}_i$, $\boldsymbol{p}_i$ and $\boldsymbol{w}_j$ respectively, where $\boldsymbol{v}_i \in \boldsymbol{M}_{\mathcal{I}}$, $\boldsymbol{p}_i \in \boldsymbol{P}$, and $\boldsymbol{w}_j \in \boldsymbol{W}$. We have $\boldsymbol{h}_i = \boldsymbol{v}_i + \boldsymbol{p}_i + \boldsymbol{w}_j$.

Basket Representation. For aggregating a set of items as a basket B^u_t, we transform item embeddings within the same basket into a basket representation by the *max pooling*. After all the items within the same basket in the item-level sequence are aggregated into a basket representation, a basket-level representation sequence is formed according to the timestamp of baskets. The basket sequence also provides essential information supplement to the item sequence.

Sorting items within a basket. We assume that the user's thinking process could be viewed as a sequential process when purchasing items. Hence, the item order within a basket is a key factor as we should expand a set of items to a list for the generative paradigm. There are different item sorting strategies to simulate user's decision process, such as sorting by (a) *item ID* (it may have implicit orders), (b) *item popularity* (popular items are more likely to be the first to think of), (c) *price* (costs are usually proportional to importances), and (d) *similarity to the last item* (similar items usually have connections). Specifically, the item popularity is calculated according to the frequency of items in the training data and then sort them from high to low. The price strategy is sorting by price from high to low. In item similarity strategy, we select the most popular item as the first one, and then add the most similar item to the last item in the basket when sorting all items. Overall, different item sorting strategies have large impacts on the results of our generative model. In this work, we adopt item popularity as the final item sorting strategy (shown in section 3.3).

Item- and basket-level encoders. The item- and basket-level encoders aim to better capture user's multi-granularity preferences by stacking multiple Transformer blocks. A Transformer block consists of two sub-layers, *i.e.,* a multi-head self-attention and a point-wise feed-forward network (FFN). The Transformer blocks in item- and basket-level encoders are the similar structures with BERT4Rec [24]. We adopt bidirectional self-attention in the encoders since it is beneficial to incorporate context from both directions for sequence

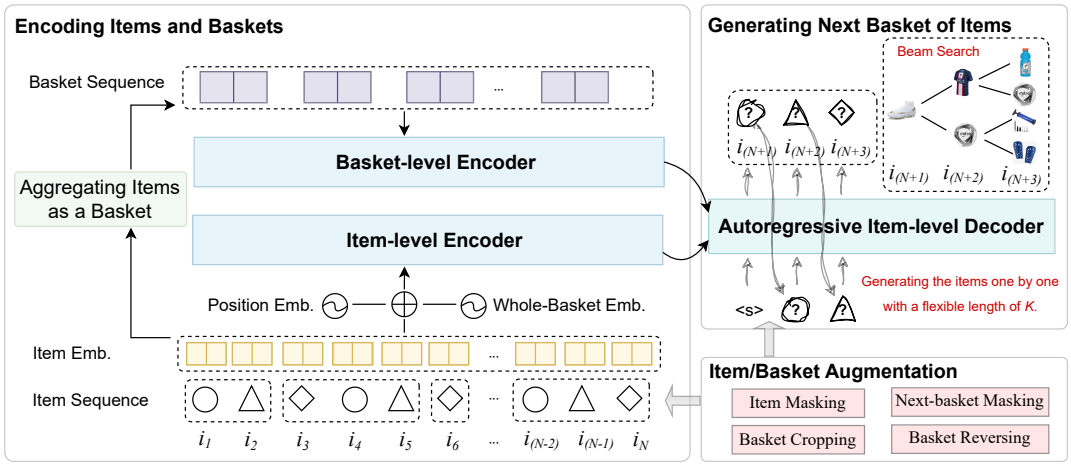

Figure 2: The overall framework of the proposed GenRec.

representation learning [6, 12, 26]. The item and basket sequences are sent into item- and basket-level encoders respectively to obtain the item-level sequence latent representation H_I^u and the basket-level sequence latent representation H_B^u, where $H_I^u \in \mathbb{R}^{N\times D}$ and $H_B^u \in \mathbb{R}^{T\times D}$. It is novel and beneficial to jointly conduct both item- and basket-level encoders, which enables a more comprehensive understanding of user sequential behavioral patterns in NBR.

Autoregressive item-level decoder. The autoregressive item-level decoder aims to generate next basket of items one by one, composed of three sub-layers, *i.e.,* a masked multi-head attention, a cross multi-head attention, and a point-wise FFN (similar structures with the decoder of BART [16]). The masked multi-head attention considers only the first $(k-1)$ items when generating k-th latent representations. The cross multi-head attention is calculate separately with item- and basket-level encoders. The *Query* in the cross multi-head attention is from the decoder and the *Key* and *Value* is from item-/basket-level encoder. We use the latent representations H_I^u,H_B^u respectively with the latent representation of masked multi-head attention to calculate the cross multi-head attention, and then two above outputs of cross multi-head attention are added. After the point-wise FFN, we can gain the decoder last hidden state $h_k^u \in \mathbb{R}^D$. With h_k^u, we can calculate the item probability distribution via softmax. To find the most probable item sequence in next basket, we utilize beam search algorithm [8] to generate next basket of items. The generative length of next basket is flexible by the hyperparameter K. Then, these K items are automatically assembled to be the next basket.

2.2 Details of Model Training

We jointly train bidirectional item- and basket-level encoders and an autoregressive item-level decoder with teacher forcing. The input of item-level encoder is the corresponding items in the basket sequence $B^u_{1,(T-1)}$, and the input of autoregressive item-level decoder is the corresponding items in the last basket B^u_T prepended with a special start token <s> and removed the last item from. The encoders and decoder are optimized to reconstruct the original basket sequence $B^u_{1,T}$. For example:

Basket Sequence: $\{B_1^u(i_1,i_2), B_2^u(i_3,i_4,i_5), B_3^u(i_6), B_4^u(i_7,i_8,i_9)\}$,
Item-level Encoder Input: $\{B_1^u(i_1,\texttt{[mask]}_1), B_2^u(i_3,i_4,\texttt{[mask]}_2), B_3^u(i_6)\}$,
Decoder Input: $\{\texttt{<s>}, B_4^u(i_7,i_8)\}$,
Labels: $\texttt{[mask]}_1 = i_2$, $\texttt{[mask]}_2 = i_5$.

Item/Basket Augmentation. We generate more training samples as item/basket augmentations to train the model. We make four transformations, which are (a) *item masking* (similar with BERT4Rec [24]), (b) *next-basket masking*, (c) *basket cropping* (*i.e.,* cropping the baskets between the start index and given end index), and (d) *basket reversing* (inspired by the help of future data for training of recommendation models [36]) as follows. These item/basket augmentations help alleviate data sparsity and bridge the gaps between item and basket sequential modeling in GenRec.

(a) **Item-level Encoder**: $\{(i_1,i_2),(i_3,i_4,i_5),(i_6)\} \xrightarrow{\text{randomly mask}} \{(i_1,\texttt{[mask]}_1),(i_3,i_4,\texttt{[mask]}_2),(i_6)\}$,
Decoder: $\{(i_7,i_8,i_9)\} \longrightarrow \{\texttt{<s>},(i_7,i_8)\}$.

(b) **Item-level Encoder**: $\{(i_1,i_2),(i_3,i_4,i_5),(i_6)\} \xrightarrow{\text{remain unchanged}} \{(i_1,i_2),(i_3,i_4,i_5),(i_6)\}$,
Decoder: $\{(i_7,i_8,i_9)\} \xrightarrow{\text{generate next basket of items}} \{\texttt{<s>}\}$.

(c) **Item-level Encoder**: $\{(i_1,i_2),(i_3,i_4,i_5),(i_6)\} \xrightarrow{\text{crop baskets}} \{(i_3,i_4,i_5),(i_6)\}$,
Decoder: $\{(i_7,i_8,i_9)\} \longrightarrow \{\texttt{<s>},(i_7,i_8)\}$.

(d) **Item-level Encoder and Decoder**: $\{(i_1,i_2),(i_3,i_4,i_5),(i_6)\},\{(i_7,i_8,i_9)\}$
$\xrightarrow{\text{reverse baskets}} \{(i_7,i_8,i_9),(i_6),(i_3,i_4,i_5)\},\{\texttt{<s>},(i_1)\}$.

Training and Inference. In training, we use the basket sequences obtained by the above four transformations as the input of encoders and decoder, and then predict the original IDs of the masked items based on the output representations of the encoders. Given a basket sequence $B^u_{1,T}$ of a user u, we define the probability of predicting an item i in the basket B^u_t as:

$$\mathrm{P}(i \in B_t^u | \widetilde{B}_{1,T}^u) = \mathrm{softmax}(\mathrm{GeLU}(h_i^u W + b_1)\boldsymbol{M}_I^{\mathrm{T}} + b_2), \quad (1)$$

where $\widetilde{B}^u_{1,T}$ is the masked basket sequence, W is the learnable projection matrix, b_1,b_2 are are bias terms, $\boldsymbol{M}_{\mathcal{I}} \in \mathbb{R}^{|\mathcal{I}|\times D}$ is the embedding matrix for the item set $\mathcal{I}$ and h^u_i is the hidden vectors of item i. To effectively learn from the training data, we adopt the negative log-likelihood of the masked items as the optimization objective:

$$\mathcal{L} = -\frac{1}{|\hat{B}^u_{1,T}|} \sum_{v_i \in \hat{B}^u_{1,T}} \log \mathrm{P}(v_i = v_i^* | \widetilde{B}^u_{1,T}), \tag{2}$$

where $\hat{B}^u_{1,T}$ is random masked items in $\widetilde{B}^u_{1,T}$, and v_i^* is the ground truth for the masked item i.

In inference, we first encode the historical basket sequence via the item- and basket-level encoders, and generate the recommended next basket of items via the autoregressive item-level decoder. Note that the token <s> acts as the input of decoder, and the length of generative next basket is controlled by the hyperparameter K. Then, we choose the items with the path of highest probability among the multiple paths generated by the beam search algorithm.

3 EXPERIMENTS

3.1 Experimental Setup

Datasets. We conduct experiments on three real-world recommendation datasets: TaFeng[1], ValuedShopper[2], and Electronics [18], a subset of Amazon dataset. For each dataset, we group interactions by the timestamp, and then generate a chronological basket sequence for each user, in which items co-purchased at the same timestamp by a user are regarded as a basket. Based on the sorted baskets, we leverage the leave-one-out strategy for each basket sequence, in which the last basket is used as the test set, the basket before the last one is used as the validation set, and the remaining baskets are used for training. The statistical details of the pre-processed datasets are shown in Table 1.

Table 1: Statistics of the pre-processed datasets. "Avg. *len*" denotes the average length of basket sequences. "Avg. *s*" denotes the average basket size.

Dataset	#Users	#Items	#Categories	#Baskets	Avg. *len*	Avg. *s*
TaFeng	12,798	11,352	2,007	73,205	5.72	6.19
ValuedShopper	8,775	6,131	821	212,267	24.19	8.26
Electronics	107,375	46,741	774	443,459	4.13	2.09

Evaluation Metrics. For the performance evaluation, we utilize recall (Recall) and normalized discounted cumulative gain (NDCG) on top-K ranked items ($K \in \{5, 10\}$), which are widely used metrics in the NBR [11, 20]. Similar with [11], we rank all items in test sets instead of sampling evaluation. We also adopt the *category-level diversity* [3, 19, 29, 31] with detailed cases to verify GenRec's capability against the over-homogenization issue. The category-level diversity (Diversity@K) is the average de-duplicated category number of top-K ranked items. The higher Diversity@K indicates that the recommendation is less homogeneous (note that item categories are only used for diversity evaluation).

[1]https://www.kaggle.com/chiranjivdas09/ta-feng-grocery-dataset
[2]https://www.kaggle.com/c/acquire-valued-shoppers-challenge/overview

Baselines and Implementation Details. We compare the proposed method with the following baseline methods: (1) conventional methods: FPMC [22], TIFU-KNN [11]; and (2) deep learning methods: BERT4Rec [24], DREAM [34], Beacon [15], Sets2Sets [10], CLEA [20],BART4Rec* [12]. For the baseline methods, we implement them with reference to their official versions if provided by the authors. Note that we adopt top-K ranked items as next basket based on the user's historical item sequence in BERT4Rec, and BART4Rec* redesigned for NBR is the modified version of E-BART4Rec [12] used for sequential recommendation. For the proposed method, We implement it[3] based on an open-source recommendation framework RecBole [37] in PyTorch, and tune the mask proportion ρ including *item masking* and *next-basket masking* using the validation set ($\rho \in \{0.15, 0.2, 0.3\}$), the length K of next basket ($K \in \{5, 10, 20\}$), and the beam size S in beam search algorithm ($S \in \{2, 5, 10\}$). For a fair comparison, we set the embedding size to 64 and batch size to 2,048. All trainable parameters in these models are optimized by the Adam optimiser with the learning rate of {0.0005, 0.001, 0.005, 0.01}, drop rate from {0, 0.1, 0.2}, and ℓ_2 regularizer from {0.001, 0.0001}.

3.2 Overall Performance on Next Basket Recommendation

We compare the proposed GenRec with the baseline methods on three real-world datasets. The results are reported in Table 2. We can make the following observations: (1) FPMC is the weakest baseline, since it only takes into account adjacent baskets, ignoring the informative long-term user preference. (2) TIFU-KNN is a strong baseline especially for NBR scenarios with lots of repeated baskets and popular items, since it can well capture the repeated purchase patterns via PIF and KNN. However, it performs worse in more diversified NBR datasets. (3) CLEA is also a strong baseline, as it denoises irrelevant items in the history and learn a more effective sequence representation. However, it also struggles with the over-homogenization issue. (4) BERT4Rec and BART4Rec* are competitive, since they can effectively obtain contextual information by the bidirectional self-attention and Cloze objective, which is confirmed in [6, 12, 24]. (5) In general, our proposed GenRec achieves the best performance in all datasets with the relative improvements ranging from 1.58% to 16.21%, demonstrating its superiority to the baseline methods. It also outperforms all deep learning baselines on "Instacart" (having more repeated baskets). (6) We attribute the improvements to three key factors: Firstly, we consider the essential correlations among items in the next basket by generating the recommended items one by one, highlighting item correlations for better surprising but reasonable item combinations in generating baskets. Secondly, we take into account both item- and basket-level context to model the user preference more effectively. Thirdly, We generate more training data to improve the model's generalization via item/basket augmentation.

3.3 Detailed Model Analyses on GenRec

Ablation Study. In this part, we analyze how different components of GenRec affect the final performance. We compare four ablation

[3]The code is available at https://github.com/RUCAIBox/GenRec.

Table 2: Overall performance of different NBR methods. The best and second-best performance methods are denoted in bold and underlined fonts, respectively. "*" indicates that the improvements are significant over all baseline methods (t-test with $p < 0.05$).

Dataset	Metric	FPMC	TIFU-KNN	BERT4Rec	DREAM	Beacon	Sets2Sets	CLEA	BART4Rec*	GenRec	Improv.
TaFeng	Recall@5	0.0639	0.0851	0.0876	0.0945	0.0837	0.0953	0.1269	0.1184	**0.1372***	+8.12%
	NDCG@5	0.0582	0.0774	0.0807	0.0834	0.0764	0.0897	0.1248	0.1132	**0.1297***	+3.93%
	Recall@10	0.0761	0.1262	0.1129	0.1182	0.1068	0.1362	0.1564	0.1421	**0.1682***	+7.54%
	NDCG@10	0.0621	0.0972	0.1064	0.1013	0.0941	0.1084	0.1457	0.1367	**0.1593***	+9.33%
ValuedShopper	Recall@5	0.0584	0.1369	0.0724	0.0667	0.0569	0.0814	0.1065	0.1354	**0.1591***	+16.21%
	NDCG@5	0.0633	0.1402	0.0893	0.0843	0.0751	0.1021	0.0979	0.1337	**0.1624***	+15.83%
	Recall@10	0.0947	0.2014	0.1082	0.0981	0.0849	0.1237	0.1621	0.2029	**0.2190***	+7.94%
	NDCG@10	0.1164	0.2136	0.1307	0.1235	0.1095	0.1564	0.1608	0.2059	**0.2287***	+7.07%
Electronics	Recall@5	0.0174	0.0295	0.0353	0.0324	0.0309	0.0291	0.0346	0.0342	**0.0361***	+2.27%
	NDCG@5	0.0098	0.0172	**0.0206**	0.0185	0.0174	0.0182	0.0197	0.0193	0.0204	-0.97%
	Recall@10	0.0209	0.0347	0.0438	0.0410	0.0398	0.0339	0.0442	0.0428	**0.0449**	+1.58%
	NDCG@10	0.0127	0.0224	0.0245	0.0229	0.0217	0.0217	0.0251	0.0244	**0.0256**	+1.99%

versions, including (1) w/o BL without basket-level encoder, (2) w/o IL without item-level encoder, (3) w/o BS that replaces beam search with greedy search, and (4) w/o AU without item/basket augmentation.

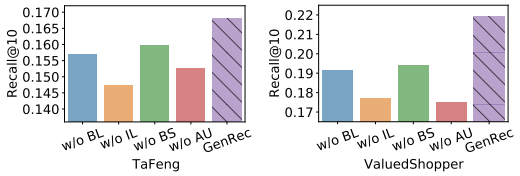

Figure 3: Ablation study of GenRec variants on "TaFeng" and "ValuedShopper". All proposed techniques have significant improvements.

The experimental results of the proposed GenRec and its variants are shown in Figure 3. We can observe that all the proposed techniques are effective to significantly improve the performance. The variants w/o IL and w/o AU have poor performances because it is essential to learning users' item-level contextual information, and more item/basket aware augmented data can effectively improve the model training for NBR.

Performance on Diversity. In Table 3, we can observe that GenRec obtains the best diversity performance, since GenRec considers item correlations of users' real purchase patterns in generating baskets via autoregressive generative decoders. TIFU-KNN considers other users' popular repeated behaviors in serving, which also has relatively good diversity (with low recall). CLEA and BERT4Rec obtain lower diversity for these mainstream deep learning based NBR methods mainly focus on the similarity between user and item representations in generating top-K items as a basket, causing the over-homogenization issue. Our generative NBR can effectively alleviate over-homogenization of predicted items in deep learning based NBR methods. More detailed analyses are in the case study part.

Table 3: Results of the category-level diversity on "TaFeng" and "Electronics".

Methods	TaFeng		Electronics	
	Recall@10	Diversity@10	Recall@10	Diversity@10
GenRec	0.1682	5.0212	0.0449	4.8824
CLEA	0.1564	4.0175	0.0442	2.9172
BERT4Rec	0.1129	3.1403	0.0438	2.0186
TIFU-KNN	0.1262	4.7994	0.0347	4.0310

Analysis on Different Item Sorting Strategies. The experimental results of different item sorting strategies within a basket are shown in Figure 4. We can observe that sorting by "item popularity" obtains the best performance, since we usually actively consider or are impressed to more popular items before long-tail items when scanning and shopping. Besides, other item sorting strategies also outperform most baselines. How to simulate user's decision process by item sorting is essential for GenRec, which will be further explored in our future works.

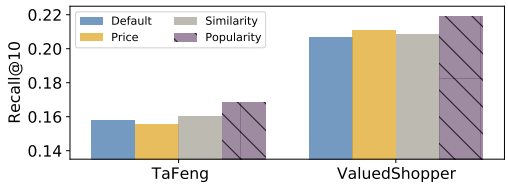

Figure 4: Results of different item sorting strategies to decide the item order in baskets.

Case Study. Table 2 and 3 quantitatively verify the power of GenRec on both accuracy and diversity. Figure 5 further gives cases for intuitive cognition. We display three typical cases given by ground truth, CLEA, and our GenRec (we conduct on Instacart for the general shopping is more intuitive and Tafeng/ValuedShopper only

have ID information). In case 1, our model provides a more balanced meal basket as ground truth compared to the homogeneous result of CLEA. In case 2, our model captures the relations among diverse daily necessities. In case 3, our model learns the preference on berries and smartly gives a more surprising *berry yogurt*, which well balances exploitation and exploration.

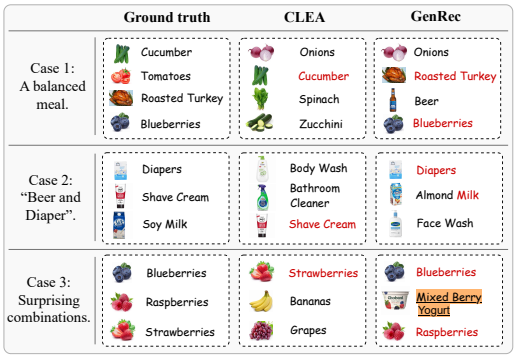

Figure 5: Examples of NBR cases. GenRec could provide more diversified, correlated, and reasonable item combinations.

4 CONCLUSION AND FUTURE WORK

In this paper, we propose a novel generative NBR model, which encodes the item- and basket-level context and generates the recommended items one by one via an autoregressive item-level decoder. Extensive experiments show that our model outperforms competitive baselines on three real-world datasets and could provide more diversified and appropriate item combinations. As future work, we will further extend the current approach to learn universal basket representation by leveraging the textual description of items, and find more accurate and practical item sorting strategy in GenRec.

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